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ANALYSIS OF IN-PLACE MATRIX ROTATION OF SQUARE MATRIX FOR INFORMATION SECURITY APPLICATIONS

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Abstract

The concept of matrices was developed to represent complex data into understandable and formattable data. There is a huge number of applications in which matrices are used like classification and data size reduction used to store it. Matrices are widely used in most cryptographic algorithms. Every image can be represented as a matrix where each pixel value is stored as an element of a matrix. The rotation of the matrix defines that a matrix is rotated in the XY-plane to a certain angle concerning the x-axis about the origin. A rotation matrix is a transformation matrix in linear algebra that is used to make a rotation in Euclidean space. The rotation of the matrix includes transformation such that it requires additional memory for a new transformed value. This paper describes the in-place matrix rotation of square matrix using 45° rotation of information. In-place algorithm alters the input without requiring additional memory. A little amount of additional memory could be needed for the implementation of an in-place algorithm. However, the amount of memory required should be consistent and not dependent on the size of the input. With the help of In-place Matrix Rotation (IMR) of the complementary angle, the IMR of 45° is proved. In IMR, the elements change their positions, but the elements remain the same in the matrix. Using IMR the matrix dimensions are changed with the center fixed to a particular position just by rotating the matrix not transforming the position.

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1. Introduction

A matrix is a rectangular array or table of numbers, symbols, or expressions organized in rows and columns that are used to represent a mathematical object or one of its properties. The horizontal and vertical lines of entries in a matrix are called rows and columns, respectively. Matrices represent linear mappings and allow for explicit linear algebra operations [1-2]. As a result, matrices are an important aspect of linear algebra, and most abstract linear algebra features and operations may be described in terms of matrices. For instance, matrix multiplication demonstrates the composition of linear maps. Matrix algebra is commonly used in geometry to express and describe geometric transformations (such as rotations) and coordinate changes [3]. Many computational issues in numerical analysis are handled by reducing them to a matrix calculation, which frequently requires computing with matrices of a huge scale. Matrices are employed in almost every branch of mathematics and science, either directly or indirectly through geometry and numerical analysis [4]. Matrix addition, scalar multiplication, transposition, matrix multiplication, row& column operations, submatrix, and rotations are some of the fundamental operations that may be used to manipulate matrices [5]. The concept of rotation is used in cryptography and it is also used in the digital image processing [6]. In the field of cryptography mostly where the block cipher are generated the matrix rotations can also be used as these blocks of data are assumed as matrices when they are creating. The rotation can be used for shuffling of data in the matrix. In the field of digital image image processing the image pixels are also considered as the matrix there the concept of rotations can be used so that the original image gets shuffled and can't be understand by others. The same methodology can be used in steganography also. By the inverse rotation the original data or pixels can be obtained again. Organizations can secure digital and analogue information using information security (InfoSec). Cryptography, mobile computing, social media, as well as infrastructure and networks carrying private, financial, and corporate information, are all covered under InfoSec. In contrast, cyber security protects both raw and useful data, but solely against internet-based dangers [7]. Information security is implemented by organisations for a variety of reasons. The primary goals of information security are often connected to safeguarding the confidentiality, integrity, and availability of firm information [8-9]. Because InfoSec encompasses so many disciplines, it frequently entails the execution of many forms of security, such as application security, infrastructure security, encryption, incident response, vulnerability management, and disaster recovery [10]. The proposed technique is made to justify that the IMR of 90° is equals to the double 45° of IMR by assuming the contradiction that the IMR of 90° is equals to the double 45° of IMR is true. It is used in the field of cryptography and digital image processing for shuffling the data or pixels in the matrix form. It is used in the block ciphers for generating the cipher text.

1.1. Some of the basic matrix operations

Matrix Addition& Subtraction

Let A, B be two matrices of size $m \times n$, then the addition of $A+B$ is given by

$$(A + B)_{i,j} = A_{i,j} + B_{i,j} \text{ where } 1 \leq i \leq m \text{ and } 1 \leq j \leq n$$

$$A - B \text{ is given by } (A - B)_{i,j} = A_{i,j} - B_{i,j} \text{ where } 1 \leq i \leq m \text{ and } 1 \leq j \leq n$$

$$A + B = B + A$$

$$A - B \neq B - A \text{ and } A - B = -(B - A)$$

Scalar Multiplication

Let A be a matrix of size $m \times n$, C is any scalar value, then the scalar multiplication is given by

$$(C.A)_{i,j} = C.A_{i,j}, \quad C \in \mathbb{Z}$$

Transposition

Let A be a matrix of size $m \times n$, then transposition of A is given by

$$(A^T)_{i,j} = A_{j,i}$$

$$((A^T)^T)_{i,j} = A_{i,j}$$

Matrix A^T will be of size $n \times m$.

Matrix multiplication

Let A be a matrix of size $m \times n$, B be a matrix of size $n \times p$, then the resulted matrix $A*B$ is of size $m \times p$ is given by

$$[AB]_{i,j} = a_{i,1}b_{1,j} + a_{i,2}b_{2,j} + \dots + a_{i,n}b_{n,j} = \sum_{r=1}^n a_{i,r}b_{r,j}$$

$$AB \neq BA$$

Representation of Linear equations as a matrix:

Matrices can be used to express and work with many linear equations, often known as systems of linear equations.

Let the system of linear equations be

$$a_1x_1 + b_1x_2 + c_1x_3 = d_1$$

$$a_2x_1 + b_2x_2 + c_2x_3 = d_2$$

$$a_3x_1 + b_3x_2 + c_3x_3 = d_3$$

Represented as: $\mathbf{Ax}=\mathbf{d}$

$$\begin{bmatrix} a_1 & b_1 & c_1 \\ a_2 & b_2 & c_2 \\ a_3 & b_3 & c_3 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} d_1 \\ d_2 \\ d_3 \end{bmatrix}$$

To get the roots of the above system of linear equations the matrices can be solved using matrix inversion method, Cramer's rule method, Gauss-Jordan Elimination Method, Triangularization Method, Cholesky Method, Partition Method, etc.,

Matrix Rotation

A rotation matrix is a transformation matrix in linear algebra that is used to make a rotation in Euclidean space [11-12]. For example, in a two-dimensional Cartesian coordinate system, the matrix rotates points in the XY plane counterclockwise through an angle θ concerning the x-axis about the origin. R is given by:

$$R = \begin{bmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{bmatrix}$$

To rotate a planar point with standard coordinates $v=(x, y)$, write it as a column vector and multiply it by the matrix R is given by:

$$Rv = \begin{bmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} x\cos\theta - y\sin\theta \\ x\sin\theta + y\cos\theta \end{bmatrix}$$

Rotation of matrices indicate rotations about the origin, since matrix multiplication does not affect the zero vector (the co-ordinates of the origin) [13]. Rotation matrices give an algebraic representation of such rotations and are widely utilized in geometry, physics, and computer graphics computations [14]. In some publications, the word rotation is broadened to cover improper rotations, which are defined by orthogonal matrices with determinants of 1 (rather than +1). These incorporate appropriate rotations as well as reflections (which invert orientation) [15]. In other circumstances, where reflections are not considered, the label proper may be omitted. Rotation matrices are real-entry square matrices [4] [13]. They are more formally orthogonal matrices with determinant 1 that is, a square matrix R is a rotation matrix if and only if $R^T = R^{-1}$ and $\det(R) = 1$. The special orthogonal group $SO(n)$ is formed by the set of all orthogonal matrices of size n with determinant +1, one example of this is a rotation group $SO(3)$ [16]. The (generic) orthogonal group $O(n)$ is formed by all orthogonal matrices of size n with determinant +1 or 1. In the Point P in Fig. 1 represents a random point in the Euclidean plane and P_1 is the transformed point of P when the object is rotated at an angle of α in the anti-clockwise direction.

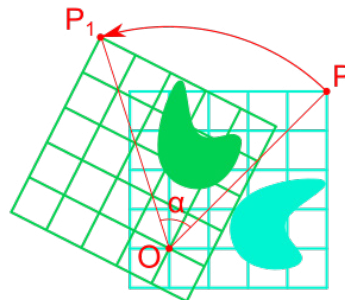


Fig. 1. Rotation of an object in 2-D around a point O through an angle α .

1.2. In-place Matrix Rotation (IMR)

Inplace matrix rotation defines the rotation of a square matrix with a certain angle around a fixed center, whereas in rotation it rotates around a fixed origin. In IMR the values in the matrix are not changed but the position of the elements is transformed. Just the view of the matrix is changed in in-place matrix rotation with the center fixed. The inplace matrix rotation can be clearly explained as assume a clock in which the hands at one end are fixed and the other is transforming its position with some angle which is observed at the fixed end. The above example is a 2D example analyze it in the 3D space then it will be easy to understand the inplace [12].

The main aim of this paper is to define In-place matrix rotation of 45° clockwise or anticlockwise with the reference of In-place matrix rotation of 90° respectively. The IMR 90° clockwise direction of the matrix is shown in fig. 2. Using this methodology, the Inplace matrix rotation of 45° is defined.

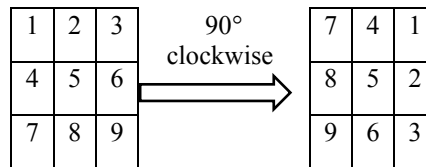


Fig. 2. In-place rotation about 90° in the clockwise direction

2. Existing Method Using 90°

The existing method deals with the In-place matrix rotation of 90° in which the size of the matrix before and after the rotation is the same, there won't be any change in the size of the matrix except that the positions of the element's changes. Let's assume the matrix as a square or rectangle and we are facing towards one side of a rectangle and rotate the rectangle such that the length becomes breadth and vice versa. This rotation of rectangle is known as in-place object rotation of 90° and the same can be applied to a matrix. In the existing technique assume the matrix is a cartesian plane and assume origin as the center of the object. A portion of the object which is placed in the first quadrant, rotates the object around the origin with a complementary angle [5]. If it is rotated in clockwise, it moves to the fourth quadrant. Similarly when it is rotated in the anti-clockwise, the portion of object moves to the second quadrant with the center as the origin. In the similar way the In-place matrix rotations can be described [12].

With the reference to the Fig. 3, the Inplace matrix rotation of 90° is done. The center of the matrix will be at the fixed position the elements around the center will rotate 90° around the center in the required direction. Figures 4 & 5 specified a total rotation of matrix by 180° ($90^\circ + 90^\circ$) is performed and the resultant matrix is equivalent to the matrix formed when 180° clockwise is performed (referenced from figure 6) on the initial matrix. This proves that combination of two complementary angle In-place matrix rotations makes one supplementary angle In-place matrix rotation. The same methodology can be applied to prove that the combination of two semi-complimentary angles can make a single complimentary angle for the in-place matrix rotation of 45° .

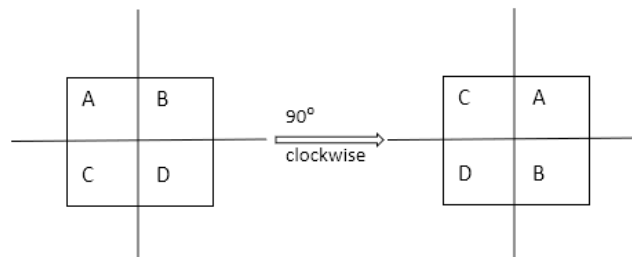


Fig 3. In-place object rotation of 90° clockwise in cartesian plane

1	2	3
4	5	6
7	8	9

90°
clockwise

7	4	1
8	5	2
9	6	3

7	4	1
8	5	2
9	6	3

90°
clockwise

9	8	7
6	5	4
3	2	1

1	2	3
4	5	6
7	8	9

180°
clockwise

9	8	7
6	5	4
3	2	1

Fig 4. IMR of 90° clockwise.

Fig 5. IMR of 90° clockwise.

Fig 6. IMR of 180° clockwise.

Assume the contradiction that the result obtained when performed Inplace matrix rotation of 45° is true. Perform one more rotation for the obtained result formed after initial Inplace matrix rotation of 45° and the formed result is customized and compared with the 90° and if both the results are equal then our assumed contradiction is true. The main reason to perform twice the rotation is that our obtained result is true by the assumed contradiction.

1	2	3	4
5	6	7	8
9	10	11	12
13	14	15	16

90° clockwise

13	9	5	1
14	10	6	2
15	11	7	3
16	12	8	4

Fig. 7. Inplace matrix rotation of 90° of a 4*4 matrix.

3. Proposed Method Using 45°

The proposed method is determining the in-place matrix rotation of 45° with the help of in-place matrix rotation of 90°. As discussed above in-place matrix rotation the name itself defines that the center of the matrix is fixed with the elements around its center rotates. The elements just change the position, but the elements aren't altered. The view of the matrix is changed.

3.1. Problem Statement: Determine value for Inplace matrix rotation of 45° in clockwise direction.

Proof: Rotate the matrix 45° with the center fixed in clockwise direction as shown in the fig. 8(a) and 8(b). the resultant matrix after rotation is shown in fig. 9.

Initial matrix

1	2	3
4	5	6
7	8	9

Fig. 8. (a) Initial matrix at 0°

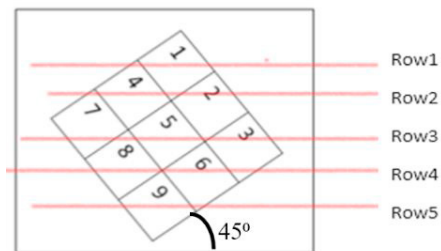


Fig. 8. (b). Inplace matrix rotation of 45°

		1		
	4		2	
7		5		3
	8		6	
		9		

Fig. 9. Resulted 5*5 matrix after the IMR of 45° is applied to the 3*3 matrix.

The 3*3 matrix after Inplace matrix rotation of 45° is changes to 5*5 matrix. *Let us assume the contradiction that the above resulted 5*5 matrix is true.* Now apply again In-place matrix rotation of 45° to the 5*5 matrix which is as shown below.

		1		
	4		2	
7		5		3
	8		6	
		9		

Fig. 10. (a) IMR of 3*3 matrix after 45°

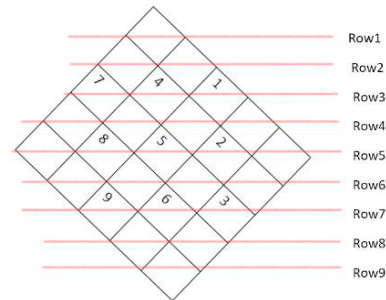


Fig. 10. (b) Inplace matrix rotation of $45^\circ+45^\circ$

The 5*5 matrix after in-place matrix rotation of 45° is changed to 9*9 matrix.

		7		4		1		
		8		5		2		
		9		6		3		

Fig. 11. Resulted 9*9 matrix after the IMR of 45° is applied to the 5*5 matrix.

But from the existing technique we know that IMR of 90° for a $n*n$ matrix result in $n*n$ matrix. With reference to this the resulted 9*9 matrix is resized to 3*3 matrix as removing the empty rows or columns which results into a 3*3 matrix and the resulted matrix is shown in the below fig 12. The main reason for removing the empty rows and columns in IMR data is that to attain 90° rotation of data, the proposed method is executed the data by two times with 45° rotation. The empty blocks are created during the implementation of 45° . To prove the contradiction true with reference the 90° rotated IMR data, the empty blocks are removed.

7	4	1
8	5	2
9	6	3

Fig. 12 Resulted 3*3 matrix after removal of empty rows and columns from the 9*9 matrix.

From fig. 4 and fig. 12 it is concluded that the result of IMR of 90° is equal to two times the IMR of 45° i.e., $(45^\circ+45^\circ=90^\circ)$.

By the above result the assumed contradiction is true. Hence the result obtained for the IMR of 45° is true and proved by the above contradiction.

3.2. Determining value for Inplace matrix rotation of 45° in anti-clockwise direction

Rotate the matrix by 45° with center fixed in anti-clockwise direction as shown in fig13 (a) &13(b).

7	4	1
8	5	2
9	6	3

Fig. 13. (a) Initial matrix at 0°

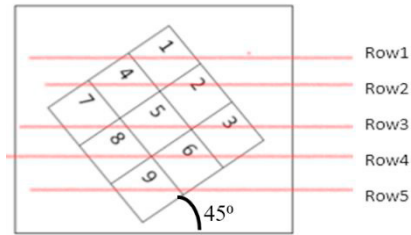


Fig. 13. (b) Inplace matrix rotation of 45°

		1		
	4		2	
7		5		3
	8		6	
		9		

Fig. 14. Resulted 5×5 matrix after the IMR of 45° is applied to the 3×3 matrix.

The 3×3 matrix after Inplace matrix rotation of 45° is changes to 5×5 matrix.

Let us assume the contradiction that the above resulted 5×5 matrix is true.

Now apply again Inplace matrix rotation of 45° to the 5×5 matrix which is as shown below.

		1		
	4		2	
7		5		3
	8		6	
		9		

Fig. 15. (a) IMR of 3×3 matrix after 45°

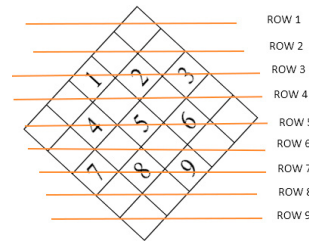


Fig. 15. (b) Inplace matrix rotation of $45^\circ + 45^\circ$

The 5×5 matrix after Inplace matrix rotation of 45° is changes to 9×9 matrix.

		1		2		3		
		4		5		6		
		7		8		9		

Fig. 16. Resulted 9×9 matrix after the IMR of 45° is applied to the 5×5 matrix.

But from the existing technique we know that IMR of 90° for a $n \times n$ matrix result in $n \times n$ matrix. With reference to this the resulted 9×9 matrix is resized to 3×3 matrix as removing the empty rows or columns which results into a 3×3 matrix and the resulted matrix is shown in the below fig. 17.

1	2	3
4	5	6
7	8	9

Fig. 17. Resulted 3*3 matrix after removal of empty rows and columns from the 9*9 matrix.

From fig 4 and fig 12 it is concluded that the result of IMR of 90° is equal to two times the IMR of 45° i.e., $(45^\circ + 45^\circ = 90^\circ)$.

From the above result the assumed contradiction is true. Hence the result obtained for the IMR of 45° is true and proved by the above contradiction.

4. Results

Operation	Initial Matrix	Matrix After IMR of 45°																																																																	
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5. CONCLUSION

The role of matrices plays a very important for various science and technological applications. The invention of matrices makes easier to represent various sets of data into simpler and understandable format. In most of the cryptographic algorithm's matrices plays an important role. For generating cipher text the data should be scrambled with the use of matrix operations. Among many matrix operations Inplace matrix rotation is one of the kind. As matrices require additional memory for rotation of data in a matrix which can be resolved with the usage of in-place matrix rotations which doesn't require additional memory for this operation? In this paper angles of multiples of 45° are only defined with the help of contradiction to 90° has proved with the algorithm. These data rotations may be applicable to secured data transmission in various technological fields.

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